

2024 Bi H2 Q11

Section: Sustainability and Interdependence

Topic: Crop Protection

Question Summary

This investigation studied the effect of the pesticide Sumithion on fish egg hatching success. Students had to identify the dependent and independent variables, describe control measures, predict and interpret a graph, and state another disadvantage of pesticide use.

Worked Solution

- (a)(i) Independent variable: concentration of Sumithion (mg/L).
- (a)(ii) Dependent variable: percentage of eggs hatched.
- (a)(iii) Controlled variables: temperature, pH, oxygen concentration, number of eggs, species of fish, and exposure time
 - all must be kept constant for valid comparison.
- (b)(i) Graph description: as Sumithion concentration increases, the percentage of eggs hatched decreases sharply. At low concentrations, hatching remains high; above a threshold (for example, 0.6 mg/L), hatching rate falls steeply towards zero.
- (b)(ii) The data suggest that Sumithion is toxic to developing fish embryos. High pesticide concentrations inhibit normal development or cause embryo death, reducing hatching success.

- (c) Other problems with pesticide use include: accumulation of residues in food chains (bioaccumulation), persistence in the environment, toxicity to non-target species, and development of resistant pest populations.

Final Answer

- > (a)(i) Independent variable - Sumithion concentration
- > (a)(ii) Dependent variable - percentage hatched
- > (a)(iii) Controls - temperature, pH, oxygen, egg number, species, time
- > (b)(i) Increasing concentration -> lower hatching percentage
- > (b)(ii) Pesticide is toxic to fish embryos
- > (c) Problems - bioaccumulation, non-target toxicity, resistance, persistence

Revision Tips

- Always identify the independent, dependent, and controlled variables clearly.
- Describe graph trends qualitatively (increasing, decreasing, constant).
- Link observed effects to biological reasoning (toxicity, interference with enzymes, etc.).
- Mention both environmental and biological effects when asked about disadvantages of pesticides.